

# Answers learning task 14

2026 – 2027

**Exercise 8.29:** A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of  $\bar{x} = 30$  mL and a standard deviation of  $s = 0.5$  mL. Find a 95 % CI on the mean volume of syrup dispensed.

Answer

[29.7936, 30.2064]

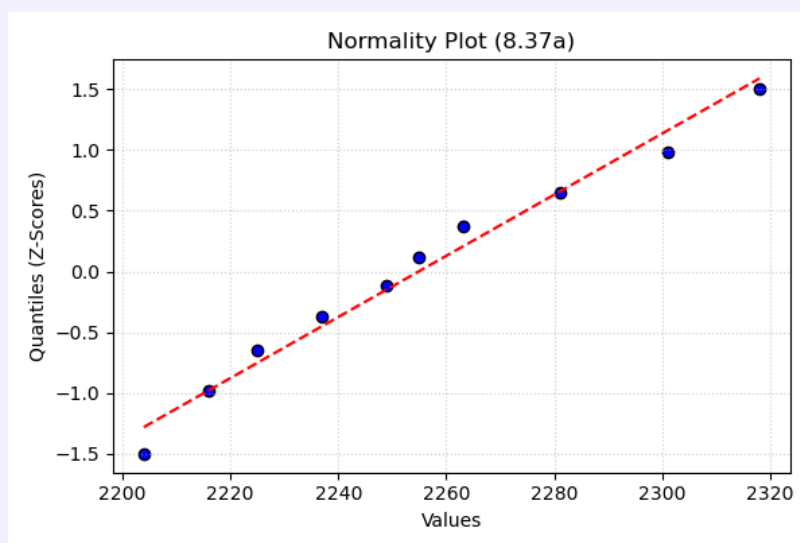
**Exercise 8.37:** The compressive strength of concrete is being tested by a civil engineer who tests 10 specimens and obtains the following data:

|      |      |      |      |
|------|------|------|------|
| 2216 | 2237 | 2249 | 2204 |
| 2225 | 2301 | 2281 | 2263 |
| 2318 | 2255 |      |      |

- (a) Check the assumption that compressive strength is normally distributed. Include a graphical display in your answer.
- (b) Construct a 95 % two-sided confidence interval on the mean strength.
- (c) Construct a 95 % lower confidence bound on the mean strength. Compare this bound with the lower bound of the two-sided confidence interval and discuss why they are different.

Answer

(a) Normal probability plot:



The points lie reasonably close to a straight line, indicating that the compressive strength data can be assumed to be normally distributed.

(b) [2228.5565, 2281.2435]

(c) [2233.5527,  $\infty$ )

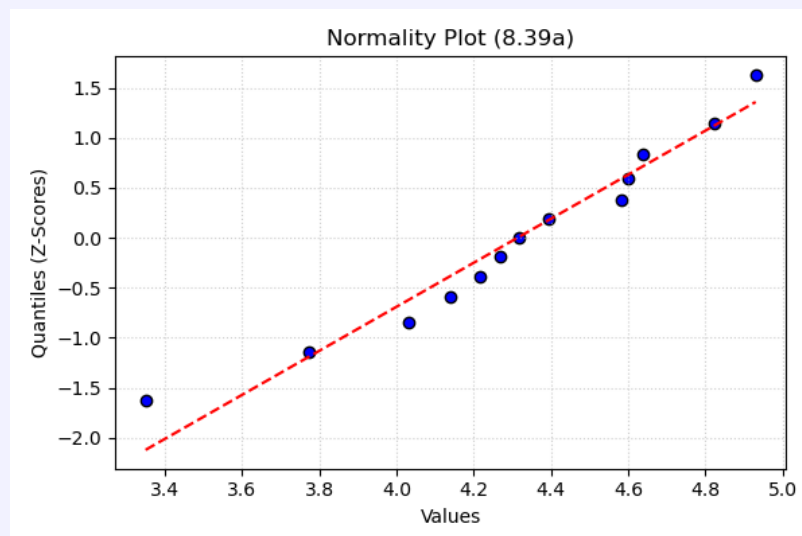
**Exercise 8.39:** An article in *Computers & Electrical Engineering* [“Parallel Simulation of Cellular Neural Networks” (1996, Vol. 22, pp. 61–84)] considered the speedup of cellular neural networks (CNN) for a parallel general-purpose computing architecture based on six transputers in different areas. The data follow:

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| 3.775302 | 3.350679 | 4.217981 | 4.030324 | 4.639692 |
| 4.139665 | 4.395575 | 4.824257 | 4.268119 | 4.584193 |
| 4.930027 | 4.315973 | 4.600101 |          |          |

- Is there evidence to support the assumption that speedup of CNN is normally distributed? Include a graphical display in your answer.
- Construct a 95 % two-sided confidence interval on the mean speedup.
- Construct a 95 % lower confidence bound on the mean speedup.

### Answer

(a) Normal probability plot:



The points approximate a straight line, showing that the speedup is normally distributed.

(b) [4.0517, 4.5748]

(c) [4.0993,  $\infty$ )

**Exercise 8.41:** An article in *Nuclear Engineering International* (February 1988, p. 33) describes several characteristics of fuel rods used in a reactor owned by an electric utility in Norway. Measurements on the percentage of enrichment of 12 rods were reported as follows:

|      |      |      |      |      |      |
|------|------|------|------|------|------|
| 2.94 | 3.00 | 2.90 | 2.75 | 3.00 | 2.95 |
| 2.90 | 2.75 | 2.95 | 2.82 | 2.81 | 3.05 |

- (a) Use a normal probability plot to check the normality assumption.
- (b) Find a 99 % two-sided confidence interval on the mean percentage of enrichment. Are you comfortable with the statement that the mean percentage of enrichment is 2.95%? Why?

Answer

- (a) The data appear to be normally distributed.
- (b)  $[2.8126, 2.9907]$ , yes, because  $2.95 \in \text{CI}$ .